MIL-F-20670B(SHIPS)
26 September 1961
SUPERSEDING
MIL-F-0020670A(SHIPS)
28 October 1958
MIL-F-20670
19 September 1952

MILITARY SPECIFICATION

FLANGES, PIPE, CARBON STEEL,

150 P.S.I., W.S.P.

(FOR NAVAL SHIPBOARD USE)

1. SCOPE

1.1 Scope.- This specification covers carbon steel pipe flanges, 150 pounds per square inch (p.s.i.), W.S.P.

1.2 Classification.- Flanges shall be of the following types, as specified (see 6.2):

Type A - Plain (slip-on).
Type B - Welding neck.
Type C - Flat-faced (slip-on).
Type D - Flat faced welding neck.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY
MIL-V-3 - Valves, Fittings and Flanges (Except for Systems Indicated herein); Packaging of.

STANDARDS

FEDERAL
FED-STD-151 - Metals; Test Methods.

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue 33rd Street, New York 16, New York).

3. REQUIREMENTS

3.1 Material.- Flanges shall be forged or machined from steel, manufactured by the open hearth, crucible, or electric furnace process.

3.2 Chemical composition and physical properties.- Flanges shall be normalized and, after normalizing, shall conform to the chemical composition and physical properties shown in tables I and II.

3.2.1 The necessary machining operations shall be performed after normalizing.

3.3 Dimensions and tolerances.-

3.3.1 Dimensions.- The dimensions of the flanges shall be shown in table III, as specified (see 6.2).

Table I - Chemical composition.

Carbon (maximum)	Manganese (maximum)	Phosphorus (maximum)	Sulfur (maximum)	Silicon (maximum)	Copper (maximum)	Nickel (maximum)	Chrome (maximum)	Molybdenum (maximum)
Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent
0.35	0.91	0.05	0.05	0.30	0.25	0.50	0.25	0.25

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Table II - Physical Properties.

Tensile strength (minimum)	Yield point (minimum)	Elongation in 2 inches (minimum)	Reduction of area (minimum)
P.s.i.	P.s.i.	Percent	Percent
60,000	30,000	25.0	35

3.3.1.1 Tolerances. - The tolerance for the flanges shall be as shown in table IV.

3.3.1.2 The maximum and minimum measurements from edge of bore to circumference of flanges shall not differ by more than 1/16 inch for sizes 3 inches and under, nor more than 1/8 inch for sizes 3-1/2 inches and over.

Table III - Flange dimensions.

Nominal size	Actual external diameter of tubing	All types (see fig. 1 through 4)		Types A and B (see fig. 1 and 2)	Types A and C (see fig. 1 and 3)	Types B and D (see fig. 2 and 4)			
		A	D	E	G	F	H	X	Y
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches
1/4	0.540	7/16	3- 1/4	1- 1/8	0.568	---	--	----	----
3/8	.675	7/16	3- 3/8	1- 1/4	.703	---	--	----	----
1/2	.840	7/16	3- 9/16	1- 3/8	.868	0.80	0.84	1- 3/16	1- 7/8
3/4	1.050	1/2	3-13/16	1-11/16	1.078	.81	1.05	1- 1/2	2- 1/16
1	1.315	1/2	4- 1/4	2	1.343	1.07	1.32	1-15/16	2- 1/8
1-1/4	1.660	1/2	4- 1/2	2- 1/2	1.686	1.42	1.66	2- 5/16	2- 1/8
1-1/2	1.900	17/32	5- 1/16	2- 7/8	1.928	1.66	1.90	2- 9/16	2- 9/32
2	2.375	9/16	5- 9/16	3- 1/2	2.422	2.13	2.38	3- 1/16	2- 5/16
2-1/2	2.875	5/8	6- 1/8	4	2.922	2.63	2.88	3- 9/16	2- 1/2
3	3.500	5/8	6- 5/8	4- 3/4	3.547	3.26	3.50	4- 1/4	2- 7/16
3-1/2	4.000	5/8	7- 3/16	5- 3/8	4.047	3.76	4.00	4-13/16	2- 1/2
4	4.500	11/16	7-11/16	5- 7/8	4.547	4.26	4.50	5- 5/16	2- 3/4
4-1/2	5.000	11/16	8- 3/16	6- 3/8	5.047	4.67	5.00	5- 7/8	3- 1/16
5	5.563	3/4	9- 1/16	7	5.648	5.23	5.56	6- 7/16	3- 5/16
5-1/2	6.125	3/4	9- 9/16	7- 5/8	6.210	5.88	6.13	7	3- 1/4
6	6.625	13/16	10- 1/8	8- 1/8	6.710	6.19	6.63	7- 9/16	3- 5/16
7	7.625	13/16	11- 5/16	9- 1/4	7.710	7.55	7.63	8- 5/8	3- 3/4
8	8.625	7/8	12- 3/8	10- 1/4	8.710	8.15	8.63	9-11/16	3- 3/4
9	9.625	15/16	13-15/16	11- 3/8	9.710	9.15	9.63	10-13/16	3- 3/4
10	10.750	1	15	12- 1/2	10.874	10.23	10.75	12	3-13/16
11	11.750	7/8	16- 9/16	13- 5/8	11.874	---	--	----	----
12	12.750	7/8	17- 5/8	14- 3/4	12.874	---	--	----	----
14 O.D.	14.000	15/16	19- 1/8	16	14.124	---	--	----	----
15 O.D.	15.000	1	20- 1/8	17	15.124	---	--	----	----
16 O.D.	16.000	1- 1/16	21- 3/16	18- 1/4	16.124	---	--	----	----
18 O.D.	18.000	1- 1/8	23- 1/4	20- 3/8	18.124	---	--	----	----
20 O.D.	20.000	1- 1/4	25-13/16	22- 1/2	20.124	---	--	----	----

3.3.1.3 The outside diameter of the raised face and the outside diameter of the flange shall be concentric. The eccentricity of raised face with respect to bore shall not exceed 1/32 inch maximum.

3.3.2 Drilling dimensions. - Unless otherwise specified (see 6.2), flanges shall be furnished drilled as shown in table V.

3.3.2.1 Tolerance. - The tolerance for drilling dimensions shall be as shown in table VI.

3.4 Marking. - The size, heat number or manufacturer's lot code identification and Specification "MIL-F-20670" shall be legibly stamped on the rim of all flanges for identification as shown on figures 1 through 4, the letters to be 1/4 to 1/2 inch high.

3.5 Finish. - The mating surface of the flanges shall have a concentric or spiral serrated finish with 30 to 80 serrations per inch of face width, with serration depth of 0.002 to 0.010 inch. The tip of the serrations shall not exceed 0.010 inch in width.

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Table IV - Tolerances.

Nominal Size	All types (see figures 1 through 4)			Types A and C (see figures 1 and 3)	Types B and D (see figures 2 and 4)			
	A	D	E	G	F	H	X	Y
	Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch
All sizes	+1/16 -1/32	----	+1/8 -0	------	----	-----	±1/16	±1/16
3 and under	----	+1/8 -1/32	----	------	----	-----	----	----
3-1/2 and over	----	+3/16 -1/32	----	------	----	-----	----	----
4-1/2 and under	----	----	----	±0.008	+0.00 -0.03	±0.031	----	----
5 to 10, inclusive	----	----	----	±0.012	+0.00 -0.06	+0.062 -0.031	----	----
Above 10	----	----	----	± .016	----	-----	----	----

Table V - Drilling dimensions.

Nominal size	Pitch circle	Number of bolt holes	Diameter of drilled bolt holes	Pitch chord
Inches	Inches		Inches	Inches
1/4	2- 1/8	3	9/16	1.84
3/8	2- 1/4	3	9/16	1.95
1/2	2- 7/16	3	9/16	2.11
3/4	2-11/16	4	9/16	1.90
1	3- 1/8	4	9/16	2.21
1-1/4	3- 3/8	4	9/16	2.39
1-1/2	3-15/16	6	9/16	1.97
2	4- 7/16	6	9/16	2.22
2-1/2	5	6	9/16	2.50
3	5- 1/2	8	9/16	2.10
3-1/2	6- 1/16	8	9/16	2.32
4	6- 9/16	8	9/16	2.51
4-1/2	7- 1/16	10	9/16	2.18
5	7-13/16	10	11/16	2.41
5-1/2	8- 5/16	10	11/16	2.57
6	8- 7/8	12	11/16	2.30
7	10	12	11/16	2.59
8	11- 1/16	14	11/16	2.46
9	12- 3/8	14	13/16	2.75
10	13- 7/16	15	13/16	2.79
11	15	16	13/16	2.93
12	16- 1/16	18	13/16	2.79
14 O.D.	17- 3/8	19	15/16	2.86
15 O.D.	18- 3/8	20	15/16	2.87
16 O.D.	19- 7/16	20	15/16	3.04
18 O.D.	21- 1/2	22	15/16	3.06
20 O.D.	23-13/16	24	1- 1/16	3.11

Table VI - Tolerances for drilling dimensions.

Nominal size	Pitch circle	Pitch chord	Eccentricity of pitch circle with respect to bore (maximum)
Inches	Inch	Inch	Inch
All sizes	±1/64	±0.015	1/64

3.6 Workmanship. - The workmanship shall be first class in every respect. Flanges shall be smooth, flat, and free from laps, seams, and other defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Sampling for acceptance inspection. -

4.2.1 Lot. -

4.2.1.1 Lot for visual and dimensional examination. - For purposes of visual, and dimensional examination, a lot shall consist of all flanges of the same type and size presented for examination at one time.

4.2.1.2 Lot for chemical analysis. - For purposes of chemical analysis, a lot shall consist of all flanges produced from the same heat of steel. When flanges cannot be identified by heat, a lot shall consist of not more than 1,000 pounds of flanges.

4.2.1.3 Lot for tension test. - For purposes of the tension test, a lot shall consist of all flanges produced from the same heat of steel and normalized in the same charge in a batch type furnace, or under the same conditions as to temperature and time in a continuous furnace. When flanges cannot be identified by heat, a lot shall consist of not more than 1,000 pounds of flanges normalized in the same charge in a batch type furnace, or under the same conditions as to temperature and time in a continuous furnace.

4.2.2 Sampling for visual and dimensional examination. - A random sample of flanges shall be selected from each lot for examination of visual and dimensional characteristics as specified in 4.3 with lot acceptance based on the sampling requirements specified in table VII in accordance with Standard MIL-STD-105.

Table VII - Sampling for visual and dimensional examination.

Lot size (number of flanges)	Sample size (number of flanges)	Acceptance number (defective)	Rejection number (defective)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 and over	50	2	3

4.2.3 Sampling for tension test and chemical analysis. - From each lot of flanges identifiable by heat (see 4.2.1.2 and 4.2.1.3) one sample flange shall be selected for the tension test (see 4.4.1) and for chemical analysis (see 4.4.2). When the flanges cannot be identified by heat samples shall be selected for tension and chemical analysis in accordance with table VIII.

Table VIII - Sampling for tension and chemical analysis tests when the heat source is not known.

Lot size (number of flanges)	Sample number of flanges to be selected for tension tests		Sample for chemical analysis
	Type A and C	Type B and D	All types
1 to 25	1	1	1
26 to 50	3	2	2
Over 50	5	3	3

4.3 Visual and dimensional examination. - Each of the sample flanges selected in accordance with 4.2.1.1 shall be visually and dimensionally examined to verify compliance with all the requirements of this specification, not involving tests. Any flange having one or more defects shall be rejected and if the number of defective flanges in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Tests. -

4.4.1 Tension test. -

4.4.1.1 Yield point and tensile strength. - The yield point shall be determined by the drop of the beam or halt in the gage of the testing machine, or by the use of dividers at a cross head speed of not to exceed 1/8 inch per minute. The tensile strength shall be determined at a speed of head not to exceed 1-1/2 inches per minute.

4.4.1.2 Test specimens. - The tension test specimens shall conform to Standard FED-STD-151.

4.4.1.2.1 In the case of small sections which do not permit taking the type 1 specimen as specified in 4.4.1.2, the tension test specimen shall conform to Standard FED-STD-151. When type 4 specimens are used they shall be made as large as feasible.

4.4.1.3 For purposes of tension tests, the necessary extra flanges or test bars shall be provided. When test specimens are cut from a flange the specimens shall be cut tangentially from the flange section approximately midway between the inner and outer surfaces and approximately midway between the front and back faces. When it is not practicable to provide extra flanges for test purposes, or when the flanges are not large enough to provide test specimens, test blanks may be made from the billet or forging bar used in the manufacture of the flanges, provided they are given approximately the same reduction and heat-treatment as for the finished flanges.

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4.4.1.4 If any tension test specimen fails to conform to the requirements of 3.2 (table II), the lot shall be rejected.

4.4.2 Chemical analysis. - Sample flanges selected in accordance with 4.2.3 shall be separately analyzed to determine compliance with 3.2 (see table I). The sample shall be free from dirt, grit, and other foreign matter, and shall consist of not more than 2 ounces. The samples shall be forwarded to a laboratory satisfactory to the bureau or agency concerned for analysis. If the analysis of any sample does not conform to the requirements of 3.2 (see table I) the lot represented by the sample shall be rejected.

4.5 Preparation for delivery inspection. - The preservation, packaging, packing and marking shall be examined to determine conformance with Specification MIL-V-3 and section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging. -

5.2.1.1.1 Level A. - The flanges shall be preserved and packaged in accordance with the level A requirements of Specification MIL-V-3.

5.2.1.1.2 Level C. - The flanges shall be preserved and packaged in accordance with the supplier's standard practice.

5.2.1.2 Packing. -

5.2.1.2.1 Level A. - The flanges shall be packed in accordance with the level A requirements of Specification MIL-V-3.

5.2.1.2.2 Level B. - The flanges shall be packed in accordance with level B requirements of Specification MIL-V-3.

5.2.1.2.3 Level C. - The flanges shall be packed to insure carrier acceptance and safe delivery to destination in compliance with Uniform Freight Classification Rules or other carrier rules and regulations as applicable to the mode of transportation.

5.2.1.3 Marking. - In addition to any special marking required in the contract or order, marking of the packages and shipping containers shall be in accordance with Standard MIL-STD-129.)

6. NOTES

6.1 Intended use. - The flanges covered by this specification are intended for Naval shipboard use for all services, as follows:

All services	10 inches and smaller	150 psi maximum for 301°F. to 775°F. maximum
		200 psi maximum for 300°F. and below
	Over 10 inches	50 psi maximum for 301°F. to 775°F. maximum
		100 psi maximum for 300°F. and below

6.1.1 Steel flanges are to be welded to seamless or welded steel pipe or tubing made in accordance to the following specifications:

MIL-T-20157 - Tube and Pipe, Carbon Steel, Seamless.
MIL-T-20162 - Tubing, Steel, Welded (including Resistance Welded) for Oil, Steam, or Water.
MIL-T-20169 - Tubing Steel, Resistance Welded (for Oil, Steam or Water).

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6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) Flange dimensions required (see 3.3.1).
(d) Whether flanges are to be furnished other than drilled (see 3.3.2).
(e) Preservation, packaging, packing or marking requirements other than those required by 5.1 (see 5.2).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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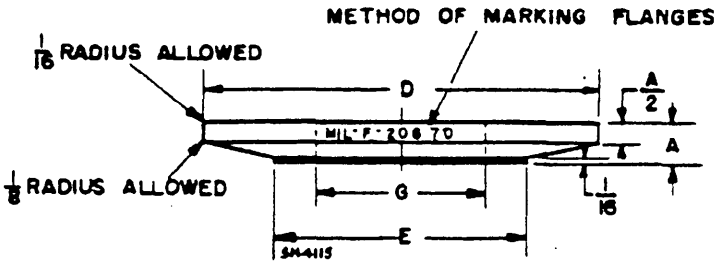


Figure 1 - Type A plain (slip-on.)

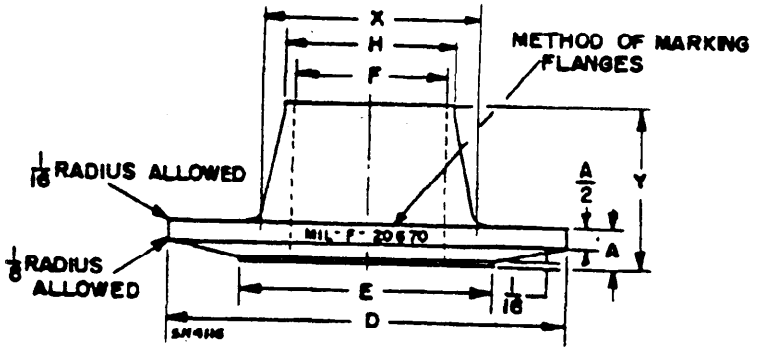


Figure 2 - Type B Welding neck

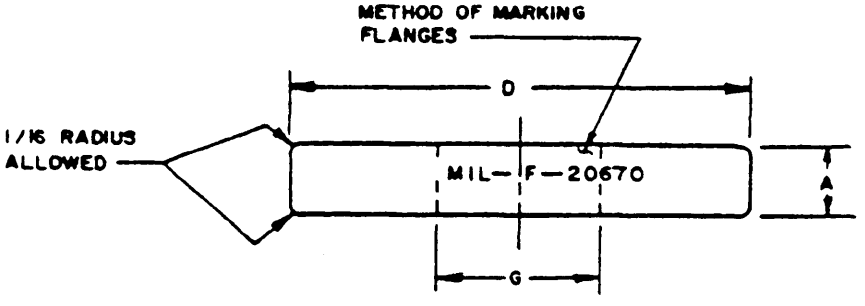


Figure 3 - Type C, flat-face (slip-on)

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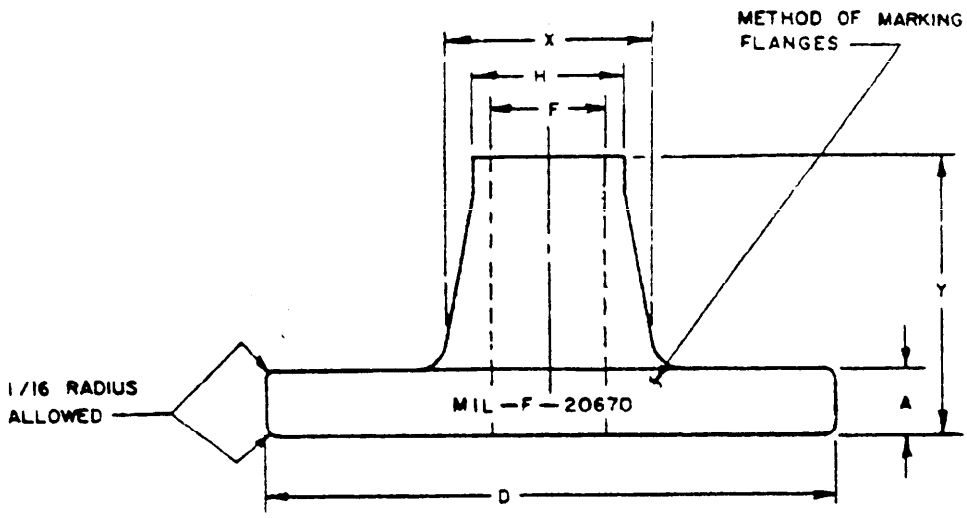


Figure 4 - Type D, flat faced welding neck

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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MIL-F-20670B(SHIPS)
AMENDMENT-2
24 April 1968

SUPERSEDING
AMENDMENT-1
29 January 1962

MILITARY SPECIFICATION

FLANGES, PIPE, CARBON STEEL

150 P. S. I., W. S. P.

(FOR NAVAL SHIPBOARD USE)

This amendment forms a part of Specification MIL-F-20670B(SHIPS), 26 September 1961.

Page 2

\# Table III: Delete column heading "X" and substitute "X[1]/" and add the following footnote at the end of the table:

"[1] Diameter of hub at base of flange."

Page 3

\# Table IV, column "H", delete and substitute:

*

H
Inch
———
———
———
-0.000 +0.031
-0.000 +0.062
-0.000 +0.062

*

\# 4.1 delete and substitute:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

Pages 7 and 8

Figures 3 and 4: Delete and substitute the attached figures.

Preparing activity:
Navy - SH
(Project 4730-N07)

[1]/ CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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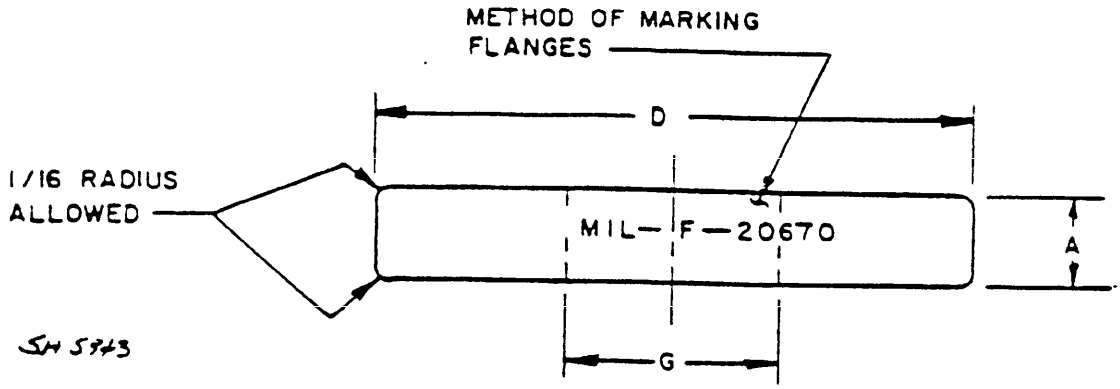


Figure 3 - Type C, flat-face (slip-on)

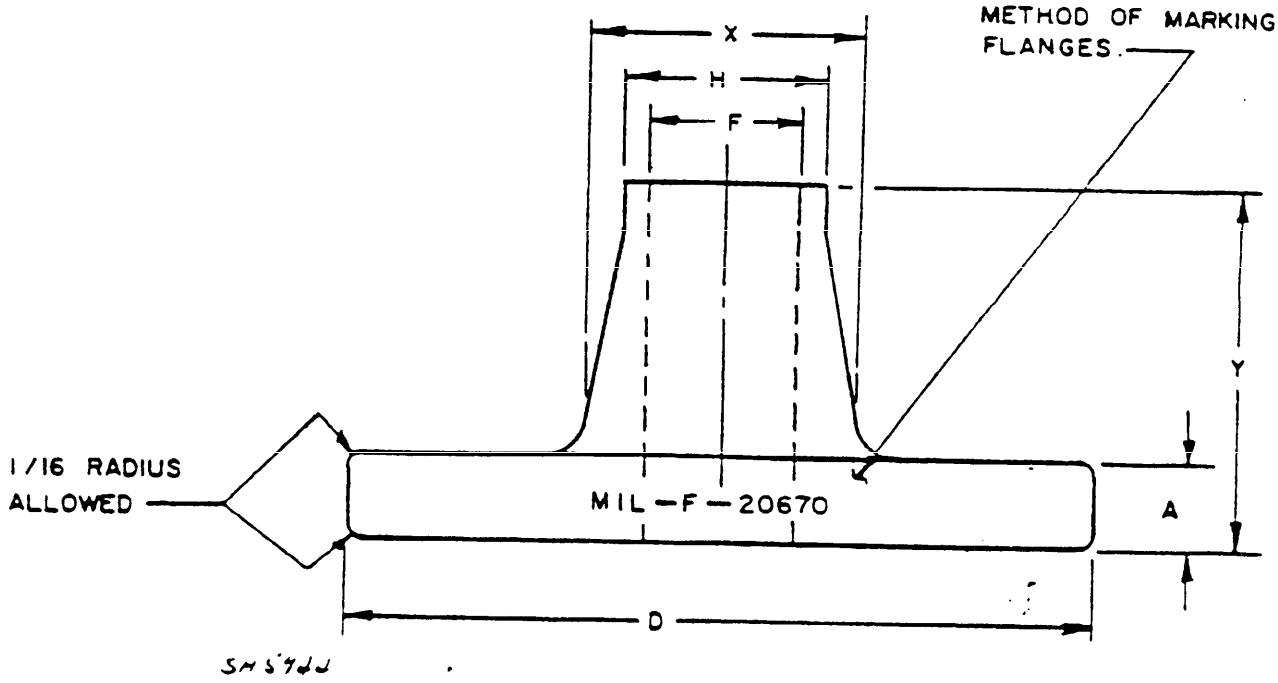


Figure 4 - Type D, flat faced welding neck"

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